ModernMolBERT: A ModernBERT Encoder Family for SELFIES Molecular Language Modeling

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Abstract

We introduce ModernMolBERT, a family of encoder-only transformer models for small-molecule representation learning. ModernMolBERT adapts the ModernBERT architecture to molecular language modelling and pairs it with a SELFIES-aware *Atom Pair Encoding* (APE) tokeniser. Both released variants (*small* and *base*) are pre-trained with masked language modelling on ∼2.4M unique SELFIES strings from ChEMBL 36 and evaluated as frozen molecular embedders on 25 binary classification benchmarks. ModernMolBERT-base attains a mean ROC-AUC of 77.4, competitive with a strong fingerprint baseline and improving on comparable pre-trained string encoders by roughly four to five points. ModernMolBERT provides a compact, reproducible, open-source foundation for SELFIES-based molecular representation learning; we release the model weights, the APE tokeniser, and training code.

1 Introduction

Molecular embeddings are the numerical interface between chemical structure and machine learning. They are the vectors passed to property predictors, virtual screening pipelines, similarity searches, clustering analyses, analogue-retrieval workflows, and dataset-triage tools. Their quality determines which chemical differences are visible to a model, which structure–activity patterns can be learned from sparse assay labels, and how well a representation transfers from known to unexplored chemical space.

Molecular representation learning is therefore an essential preprocessing step. A useful molecular embedder should preserve chemically meaningful structure, avoid artefacts introduced by molecular notation, and be efficient enough to apply across large compound collections. ModernMolBERT addresses all of these by combining chemically valid SELFIES strings, a SELFIES-aware Atom Pair Encoding tokeniser, and an efficient ModernBERT encoder trained as a reusable frozen molecular embedder (Figure 1).

For decades, the dominant answer was to map each molecule to a fixed-length vector of hand-crafted descriptors, such as physicochemical properties and substructure-based fingerprints. The most prominent examples are circular fingerprints such as the Extended Connectivity Fingerprint (ECFP) [1], which hash the presence of circular atomic neighbourhoods at increasing radii into a fixed-length bit vector and have been a workhorse of quantitative structure–activity relationship (QSAR) modelling. Their compactness and interpretability are desirable properties, but their fixed dimensionality and bespoke design mean they cannot adapt to new structural motifs, lose information through hashing collisions, and provide no mechanism for learning richer structure–activity correlations from data. An early bridge between fixed fingerprints and fully learned representations was Mol2Vec [2], which applied the Word2Vec algorithm to sequences of Morgan substructure identifiers, learning distributed embeddings for molecular fragments without any labelled data. Learnable alternatives that operate directly on molecular graphs — treating atoms as nodes and bonds as edges — have also more recently been proposed [3, 4], but these require specialised aggregation schemes and sit outside the mainstream transformer pretraining paradigm.

The BERT [5] family of bidirectional encoder models revolutionised how masked language model (MLM) pretraining on large unlabelled corpora could produce representations that transfer to many downstream tasks with minimal fine-tuning. Molecules are encoded as SMILES (Simplified Molecular-Input Line-Entry System) strings [6], a linear notation that encodes atomic connectivity as a depth-first traversal of the molecular graph. This sequential form is a natural substrate for the same sequence models originally developed for words and sentences.

SMILES molecular language models followed in rapid succession: ChemBERTa [7], MolBERT [8], ChemBERTa-2 [9], and MoLFormer [10]. These collectively established that MLM pretraining on SMILES yielded representations competitive with fingerprint baselines on property-prediction benchmarks without task-specific feature engineering, and that the approach scales to billion-molecule corpora. Table 1 summarises their key design choices alongside ModernMolBERT; Section 2 discusses them in detail.

Model	Repr.	Tokeniser	Arch.	Params	Corpus	Size (M)
ChemBERTa [7]	SMILES	BPE	RoBERTa	5–46[†]	PubChem	77
ChemBERTa-2 [9]	SMILES	BPE/atom	RoBERTa	5–46[†]	PubChem	77
MolBERT [8]	SMILES	atom	BERT	85	ChEMBL	2
MoLFormer [10]	SMILES	atom	Linear-attn.	46.8	ZINC+PubChem	>1,100
SELFormer [11]	SELFIES	BPE	RoBERTa	87	ChEMBL	2
ModernMolBERT-small	**SELFIES**	**APE**	**ModernBERT**	**34.1**	**ChEMBL 36**	**2.4**
ModernMolBERT-base	**SELFIES**	**APE**	**ModernBERT**	**114.3**	**ChEMBL 36**	**2.4**

Table 1: Overview of representative molecular string encoder models. *Repr.*: molecular string representation used for pretraining. *Tokeniser*: tokenisation strategy (BPE = Byte Pair Encoding; atom = atom-level; APE = Atom Pair Encoding). *Arch.*: encoder backbone. *Params*: trainable parameters in millions. *Size*: pretraining corpus size in millions of molecules. Embedding dimensions for these and additional models are listed in Table 8 (Section D). **Bold** rows indicate the models introduced here. [†]The ChemBERTa family spans several released checkpoints of approximately 5–46M parameters; the 77M label denotes pretraining molecules, not parameters.

SMILES-based language models have two weaknesses as notation. First, most perturbations of a SMILES string are chemically invalid [12], so a model must spend capacity recognising ill-formed inputs, and a single molecule admits many equivalent strings. This inflates the effective vocabulary and complicates self-supervised objectives. Second, subword tokenisers such as Byte Pair Encoding (BPE) [13], applied naïvely to molecular strings, split multi-character atomic symbols (e.g. `Cl` into `C` + `l`) [14], destroying the chemically meaningful units a molecular tokeniser should preserve.

SELFIES (Self-Referencing Embedded Strings) [12] addresses the validity problem by construction. Its context-free grammar guarantees that every syntactically valid SELFIES string decodes to a chemically valid molecule under any editing or sampling operation, a guarantee SMILES cannot provide (Figure 2A). It removes invalid-string noise from the pretraining signal while preserving the sequential structure that makes transformer encoders effective. Yet SELFIES has been adopted mainly for molecular *generation*, while encoder-only SELFIES representation learning remains comparatively unexplored [14]. To our knowledge the combination of SELFIES, a SELFIES-adapted atom-preserving tokeniser, and a ModernBERT-style encoder has not been systematically evaluated as a molecular embedder.

Tokenisation, the mapping from a raw string to discrete vocabulary units, is almost as essential as the model itself. Atom-level tokenisation is chemically principled but yields long sequences and misses recurring substructures, whereas BPE operates on raw character frequencies and produces entries that straddle atom boundaries. Leon et al. [14] proposed *Atom Pair Encoding* (APE), which adapts the BPE merge procedure to respect molecular structure by restricting merges to adjacent complete molecular symbols. We adapt APE to SELFIES by merging only adjacent complete SELFIES primitives, so every vocabulary entry is a valid concatenation of complete SELFIES symbols rather than a character-level artefact (Figure 2C). APE thus occupies a middle ground: compact like subword tokenisation, chemically principled like atom-level tokenisation.

Many molecular string encoders such as ChemBERTa, ChemBERTa-2, MolBERT, and SELFormer are built on BERT- or RoBERTa-era backbones [5, 15] with absolute positional embeddings and full quadratic attention. Several post-BERT advances have been consolidated in ModernBERT [16]: Rotary Position Embeddings (RoPE), Flash Attention, alternating local/global attention, and GeGLU activations (Section 2.3). To our knowledge, these have not been applied to SELFIES-based molecular language modelling. Especially RoPE may be suited to molecular strings, where absolute sequence position is partly an artefact of the chosen graph traversal rather than an intrinsic molecular property. Because RoPE conditions attention on the *relative* offset between two tokens rather than their absolute indices, the encoder becomes largely invariant to where a traversal happens to begin — precisely the component of SMILES/SELFIES positioning that carries no chemical meaning — while still distinguishing the local ordering of neighbouring symbols.

The MLM objective does not directly specify which tokens to mask. Standard BERT masks tokens independently at random [5], whereas SpanBERT [17] masks contiguous spans to encourage modelling of longer-range structure. Masking can also be made chemically informed: Peng et al. [18] mask complete functional groups identified by parsing, reasoning that functional groups carry disproportionate property-relevant information. Whether such structure-aware masking benefits a frozen SELFIES embedder is an open question we examine in an ablation (Section 5.2).

We introduce ModernMolBERT, a compact family of encoder-only molecular language models combining three design choices not previously studied together: a ModernBERT backbone, chemically valid SELFIES inputs, and a SELFIES-adapted APE tokeniser. The models are trained from scratch with MLM on 2.4M curated ChEMBL 36 molecules and released in small and base variants for molecular property prediction. We evaluate them on the 25-dataset benchmark of Praski et al. [19] against a fingerprint baseline and representative pre-trained string encoders. Figure 1 shows the pretraining and frozen-evaluation pipeline, and Figure 2 illustrates the two molecular design choices that distinguish ModernMolBERT from prior SELFIES encoders.

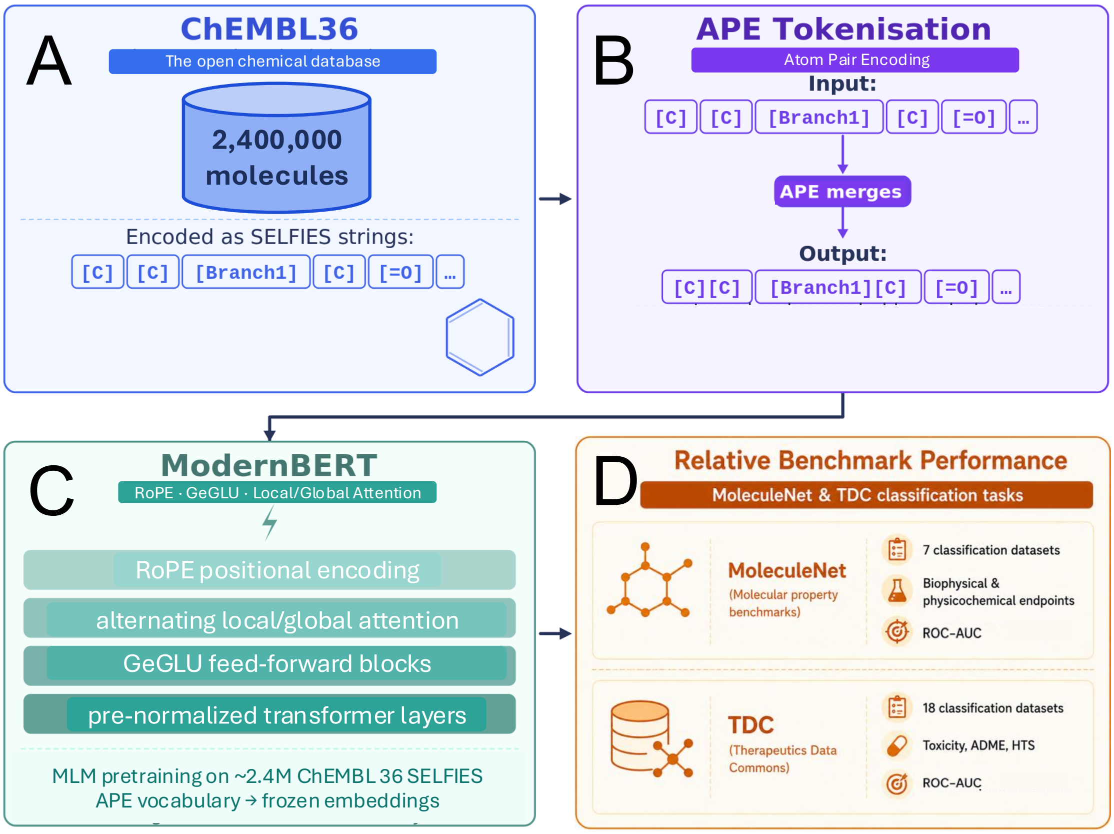

Figure 1: **ModernMolBERT pretraining and frozen-evaluation pipeline.** Approximately 2.4M unique CHEMBL 36 small molecules are curated, canonicalised, and converted to SELFIES strings, then pre-tokenised with the SELFIES-adapted APE tokeniser (Figure 2C). A MODERNBERT encoder is trained from scratch on this corpus with the masked language modelling (MLM) objective, yielding MODERNMOLBERT. For evaluation the encoder is *frozen*: mean-pooled final-layer token embeddings over non-special SELFIES tokens are extracted without any gradient updates and passed to lightweight downstream classifiers (ridge, random forest, k-nearest neighbours) on 25 binary molecular property-prediction benchmarks, where MODERNMOLBERT is compared against fixed fingerprint and pre-trained string-encoder baselines under an identical protocol.

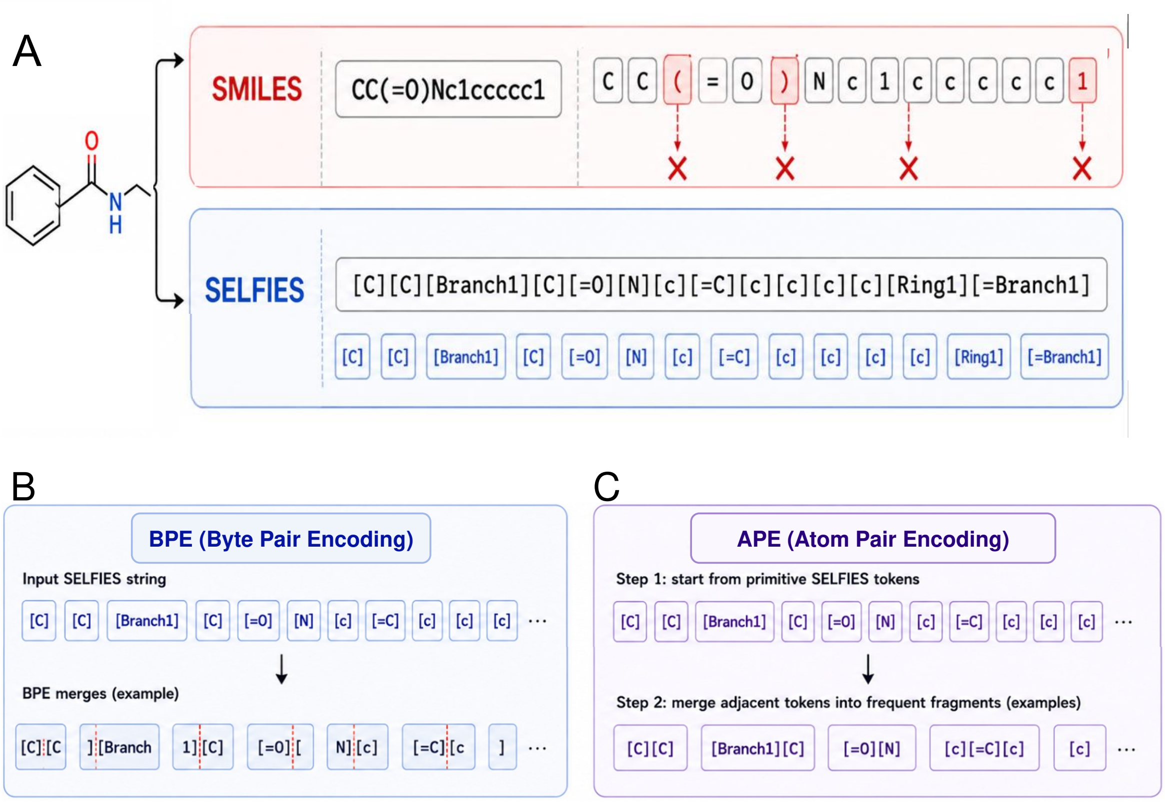

Figure 2: **Method comparison of the two molecular design choices in ModernMolBERT: input representation and tokenisation.** **(A)** *Input representation: SMILES versus SELFIES.* The same molecule is shown encoded as a SMILES string (top) and as a SELFIES string (bottom). SMILES tokens can violate chemical syntax under perturbation or naive tokenisation — mismatched brackets and ill-formed ring-closure indices are marked with red crosses — whereas every syntactically valid SELFIES *string* decodes to a chemically valid molecule [12]. **(B)** *Byte Pair Encoding (BPE).* BPE merges SELFIES symbols by raw character frequency without chemical awareness, splitting bracketed symbols and producing chemically meaningless vocabulary entries (dashed red markers). **(C)** *Atom Pair Encoding (APE).* APE restricts merges to adjacent complete SELFIES primitives, so every vocabulary entry is a valid concatenation of complete SELFIES symbols rather than a character-level artefact [14].

2 Related Work

The introduction motivated the three design choices behind ModernMolBERT; this section places them in the context of prior work along the four axes most relevant to a frozen SELFIES embedder — molecular string representations, molecular language models, modern encoder architectures, and molecular tokenisation — and identifies the specific models we later use as baselines.

2.1 Molecular String Representations

The chemical-validity guarantee of SELFIES holds under arbitrary mutation: as reported by Krenn et al. [12], a single random bit-flip leaves a SMILES string valid only 26.6% of the time and ten bit-flips reduce validity to 0.2%, whereas SELFIES remains 100% valid in both cases. The representation is also complete, meaning that every molecule can be encoded, and it can serve as a drop-in input to any model. In generative settings, VAEs and GANs trained on SELFIES populate their latent spaces with two orders of magnitude more diverse valid molecules than equivalent SMILES-based models [12].

2.2 Molecular Language Models

The fingerprint ECFP [1] remains the dominant deterministic baseline: a circular-substructure fingerprint with no trainable parameters that continues to compete with learned representations on many property prediction tasks. Among learned string encoders, ChemBERTa-2 [9] revisited the RoBERTa-on-SMILES recipe, pretraining on PubChem SMILES and comparing masked language modelling with multi-task regression (MTR) over computed RDKit descriptors. MoLFormer [10] scaled the masked-language-modelling approach to 1.1B molecules and introduced a linear-attention transformer, making it the largest-scale SMILES string model and a key large-scale reference point. SELFormer [11] is the closest prior work to ModernMolBERT: a RoBERTa encoder pre-trained with MLM on 2M drug-like SELFIES strings from ChEMBL. Several adjacent molecular foundation models use different pretraining objectives or architectures. Chemformer [20] uses a sequence-to-sequence transformer for computational chemistry tasks, whereas CDDD [21] learns continuous molecular descriptors by translating between equivalent molecular strings. Recent encoder-only work such as ChemBERTa-3 [22] and MolEncoder [23] continues to refine training recipes for chemical BERT-style models. Recent contrastive and multimodal approaches such as CLAMP [24] and COATI [25] extend molecular embeddings beyond pure string reconstruction by incorporating assay text, 3D information, or other paired supervision. These models broaden the representation-learning landscape but answer a different question from the one addressed here: how far a compact, encoder-only SELFIES MLM can be pushed as a frozen molecular embedder. A large and growing literature applies graph neural networks and graph transformers to molecular representation learning; representative examples include GROVER [4], GIN [26], GraphMVP [27], and Uni-Mol [28]. These graph-based methods are outside the scope of ModernMolBERT, which operates on sequential SELFIES strings without explicit graph structure. Broader benchmarking studies have nevertheless shown that the relative value of learned molecular representations depends strongly on the dataset, split strategy, chemical-series structure, and amount of labelled data available [29, 30]. This motivates treating ECFP4 and other fixed representations as serious baselines rather than as outdated reference points.

2.3 Modern BERT Architectures

ModernBERT [16] is an encoder-only transformer that consolidates post-BERT architectural advances into a production-quality model trained on 2 trillion tokens of text, code, and scientific literature with a native 8 192-token context window. Its key design choices are: Rotary Position Embeddings (RoPE) [31] replacing absolute positional encodings; *alternating local and global attention*, in which most layers apply a sliding-window kernel and periodically interleaved layers apply full-sequence attention, executed with the IO-aware Flash Attention [32] kernel; GeGLU feed-forward activations; and unpadded sequence processing, which removes padding tokens to eliminate wasted compute. The two released variants for natural language use are ModernBERT-base (149M parameters) and ModernBERT-large (395M). They are trained with a 30% MLM masking rate and achieve state-of-the-art results among encoder-only models on natural-language and code benchmarks.

2.4 Molecular Tokenisation

The atom-preserving APE tokeniser [14] (Section 3.1) was shown to outperform standard BPE in a controlled comparison of RoBERTa-based models pre-trained on 1M PubChem molecules and fine-tuned on MoleculeNet, with the largest gains on SELFIES representations (ROC-AUC improvements of +0.059 on BBBP and HIV and +0.017 on Tox21; Table 6 of Leon et al. 14), a result attributed to its merges aligning with SELFIES's symbol-level structure.

3 Methods

3.1 Atom Pair Encoding Tokeniser

We use a custom reimplementation of the APE algorithm [14, 33] with several modifications for SELFIES pretraining at scale, detailed in the paragraphs below: molecular-aware pre-tokenisation that restricts merges to complete SELFIES primitives, frequency- and span-controlled vocabulary construction, deterministic greedy longest-match inference, and post-training injection of benchmark-observed primitives.

Molecular-aware pre-tokenisation. SELFIES strings are first split on a regex that isolates each complete bracket-enclosed primitive, so merges operate only over whole SELFIES primitives and every vocabulary entry is a valid concatenation of complete symbols.

Vocabulary construction. Merging is controlled by two criteria: a minimum merge frequency (`min_freq` = 3000), which halts merging once the most frequent adjacent pair co-occurs fewer than 3 000 times, and a maximum merge span (`max_merge_pieces` = 2), which excludes candidates spanning more primitives to avoid long, uninformative concatenations. In practice `min_freq` is the binding early-stopping criterion. Full tokeniser statistics are reported in Table 5 (Section B).

Greedy longest-match tokenisation. At inference, each position is consumed by the longest matching vocabulary entry, looking up to `max_merge_pieces` positions ahead. The procedure is deterministic and $O(n \cdot \texttt{max\_merge\_pieces})$ per sequence, with an `<unk>` fallback only when no single primitive is in-vocabulary.

Post-training symbol injection. After merge learning but before model pretraining, any valid SELFIES primitive that appears ≥ 10 times in the benchmark molecules but is absent from the learned vocabulary is appended. This uses only primitive-frequency information; no benchmark labels or train/test split information are used in this step. It guarantees a zero unknown-token rate on evaluation molecules; all model weights were initialised and trained on the completed 631-token vocabulary from the outset.

3.2 Model Architecture

Both ModernMolBERT variants use the ModernBERT encoder backbone (Section 2.3) [16] in its default configuration: RoPE, GeGLU feed-forward blocks, pre-normalisation, no linear-layer biases. It uses unpadded Flash Attention with global attention every 3 layers and a 128-token local sliding window. They are trained from scratch on our SELFIES APE vocabulary; no weights are transferred from the official ModernBERT checkpoints [34]. Per-variant hyper-parameters are listed in Table 3 (Section A). ModernMolBERT-small (34.1M parameters, 512-dimensional hidden state) sits in the compact regime below Chemformer [20] (45M), MoLFormer-XL [10] (46.8M), and Uni-Mol [28] (47.6M), and is notably smaller than the closest SELFIES comparator SELFormer (86.7M). ModernMolBERT-base (114.3M parameters, 768-dimensional) is significantly larger than the existing models considered here.

3.3 Pretraining Data

Pretraining data are sourced from ChEMBL 36 [35] via the `lukaskim/ChEMBL-36` Hugging Face dataset[1]. Molecules are deduplicated by `standard_inchi_key`, parsed and lightly sanitised with RDKit [36] (ring perception and aromaticity assignment only, avoiding full sanitisation), and canonicalised to isomeric canonical SMILES using RDKit's `MolToSmiles` with `isomericSmiles=True`. The resulting canonical SMILES strings are then encoded to SELFIES [12] using the `selfies` Python package. Molecules that fail RDKit parsing, limited sanitisation, canonicalisation, or SELFIES encoding are excluded. We further filter by size and type, retaining only entries flagged as small molecules with $3 \leq$ heavy atoms ≤ 100 and molecular weight $\leq 1000\,\mathrm{Da}$. Train and validation splits (99%/1%) are assigned deterministically by SHA-1 hashing of a stable per-molecule identifier (preferring the standard InChIKey, with fallbacks to ChEMBL ID, canonical SMILES, and SELFIES in that order) with seed 13, yielding 2390317 training and 24228 validation molecules. The resulting SELFIES strings are subsequently pre-tokenised with the APE tokeniser [14] using greedy longest-match tokenisation. The full curation and pre-tokenisation pipeline is available at `https://github.com/HauserGroup/ModernMolBERT`. The final dataset is available on `https://huggingface.co/HauserGroup`.

3.4 Pretraining Procedure

Both ModernMolBERT variants are trained from scratch with the masked language modelling (MLM) objective and no auxiliary tasks. Training runs for 30,000 steps at an effective batch size of 256 and a maximum sequence length of 128 tokens. For each variant, the masking rate and peak learning rate are selected by a 3×3 grid search (Section 5.2); the remaining optimisation settings (AdamW, weight decay, gradient clipping, `bf16`, linear

[1] `https://huggingface.co/datasets/lukaskim/ChEMBL-36`

warmup, and cosine decay) are fixed and reported in Table 4 (Section A). Validation loss and masked-token accuracy converge smoothly within the fixed step budget (Figure 8).

Masking strategies. We implement three masking strategies as a selectable data-collator component. *Standard* masking draws an independent Bernoulli mask per eligible token. *Span* masking masks contiguous spans with lengths drawn from a Geometric distribution ($p = 0.4$, clamped to 6 tokens), following SpanBERT's span selection but without its Span Boundary Objective [17]. *Heteroatom-biased span* masking up-weights span-start positions at APE tokens containing a heteroatom primitive (N, O, S, P, F, Cl, Br, I, Se, Si), identified by a bracket-aware SELFIES regex; spans extend beyond the start, so masking is enriched around rather than restricted to these tokens. It is a parser-free proxy for the functional-group masking of Peng et al. [18], using the APE bracket vocabulary in place of explicit SMARTS matching. All three apply the standard BERT corruption rule (80% `[MASK]`, 10% random token, 10% unchanged). The released models use standard masking; Section 5.2 compares all three.

3.5 Evaluation Protocol

All benchmark tasks are evaluated using frozen ModernMolBERT embeddings, following the lightweight protocol of Praski et al. [19].[2] For each molecule, the encoder is run in inference mode and molecule-level representations are obtained by mean-pooling the final-layer token embeddings over non-special SELFIES tokens; no model parameters are updated. These embeddings are evaluated with three lightweight downstream models chosen to span complementary inductive biases — a linear ridge classifier, ensemble random forests, and instance-based k-nearest neighbours — so that representation quality is not judged through a single decision boundary. We adopt these three heads, and their cross-validation search grids, directly from the benchmark protocol of Praski et al. [19] to keep the comparison with their reported baselines exact; hyperparameters are selected by cross-validation on the training split, with performance reported on the held-out test split. All tasks are scored by ROC-AUC. For multi-task datasets, missing endpoint labels are treated as unmeasured: downstream classifiers are fit endpoint-wise on finite labels (in both cross-validation and test evaluation), and dataset-level ROC-AUC is averaged over endpoints with finite labels and both classes present in the test split. The best downstream model per dataset is used in the main comparison.

4 Experiments

4.1 Benchmark Datasets

We evaluate on 25 binary classification datasets drawn from two sources: 18 datasets from the Therapeutics Data Commons (TDC) [37], spanning toxicity (4 datasets), ADME (12 datasets), and high-throughput screening (2 datasets); and 7 datasets from MoleculeNet [38], covering a range of biophysical and physical chemistry endpoints. Dataset sizes range from 475 to 93 087 compounds, with task counts ranging from 1 to 27 (SIDER). The full list with per-dataset statistics is given in Table 4 of Praski et al. [19], whose benchmark selection we follow directly. All datasets are evaluated by ROC-AUC.

[2] Using their public benchmark code at https://github.com/MLCIL/benchmarking_molecular_models.

4.2 Baselines

We compare ModernMolBERT against both fixed and learned molecular representations. ECFP4 is the classical non-neural baseline [1, 29, 30]. The pre-trained string-model baselines each isolate one axis of comparison: ChemBERTa-2 is the closest-scale RoBERTa-style SMILES MLM; SELFormer isolates the use of SELFIES with a BERT/RoBERTa-era encoder; and MoLFormer[3] is a large-scale SMILES reference trained on substantially broader PubChem and ZINC data. Because ModernMolBERT is trained purely with MLM, we use the MLM-trained variant of each baseline so that the comparison is not confounded by the pretraining objective: SELFormer and MoLFormer are MLM-pre-trained [10, 11], and for ChemBERTa-2 we use the MLM checkpoint (`seyonec/ChemBERTa-77M-MLM`) rather than the multi-task-regression variant [9]. Auxiliary-objective and multi-task-regression pretraining therefore falls outside this MLM-only comparison. No model, including ModernMolBERT, is fine-tuned end-to-end; all neural baselines are evaluated as frozen embedders under the same downstream protocol.

5 Results

5.1 ModernMolBERT Approaches Fingerprints and Improves on Closest-Scale String Encoders

The main comparison aggregates ROC-AUC across the 25 binary classification benchmark datasets described in Section 4.1. For each representation–dataset pair, we report the best cross-validated downstream estimator among ridge classification, random forests, and k-nearest neighbours, following the frozen mean-pooled embedding protocol in Section 3.5. For multi-endpoint datasets, endpoint ROC-AUCs are averaged before aggregation. This presentation separates the quality of the pre-trained representation from gains attributable to end-to-end fine-tuning, and therefore directly tests whether ModernMolBERT provides useful off-the-shelf molecular embeddings. Throughout, ROC-AUC values are reported $\times 100$, so one point corresponds to 0.01 ROC-AUC. Direct paired comparisons use only jointly evaluated datasets, reported in Figure 3 and Table 7. Table 2 reports mean ROC-AUC by task group; per-dataset results for all models are given in Table 6.

ModernMolBERT-base attains a mean ROC-AUC of 77.4 across 24 evaluated benchmark datasets (Tox21 is omitted for ModernMolBERT-base; see Table 6), and ModernMolBERT-small attains 77.3 across all 25 datasets (per-dataset results in Table 6). For comparison, ECFP4 reaches 78.9, MoLFormer 79.8, ChemBERTa-2 72.9, and SELFormer 72.1. Both ModernMolBERT variants therefore land between the strong fixed-fingerprint and large-scale SMILES baselines (ECFP4, MoLFormer) and the closest-scale pre-trained string encoders (ChemBERTa-2, SELFormer), which they exceed by roughly four to five ROC-AUC points. Figure 4 shows the corresponding per-dataset paired comparisons against each baseline.

By task group, ModernMolBERT is strongest on the property-driven TDC tasks: it is competitive with ECFP4 on ADME (ModernMolBERT-base 80.7 vs. 81.1) and toxicity (83.8 vs. 85.0), the endpoints most closely tied to physicochemical structure (Figure 5). Its weakest group is the two-task TDC-HTS set (64.4), where all neural embedders trail ECFP4.

[3] MoLFormer's inference relies on the ONNX Runtime for its linear-attention kernel, which has limited compatibility with Python newer than $\approx$3.10; embedding extraction therefore required a pinned legacy environment.

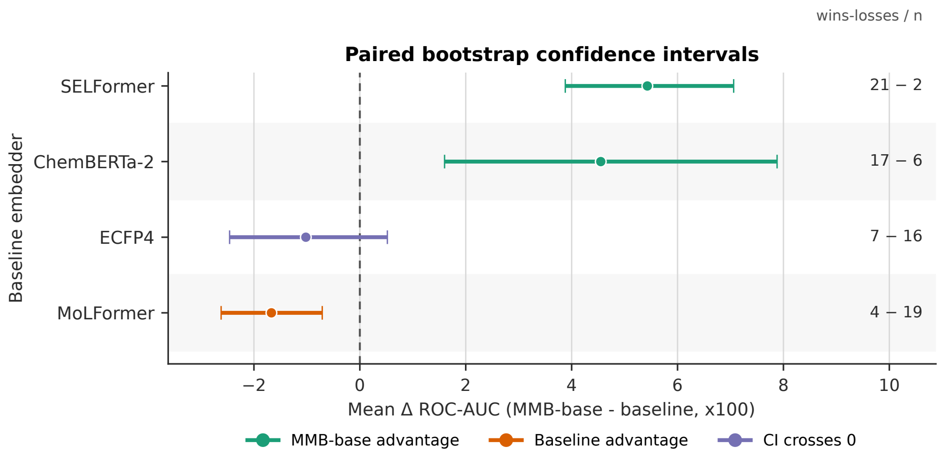

Figure 3: **Paired bootstrap confidence intervals for ModernMolBERT-base vs. each baseline ($n = 23$ jointly evaluated datasets).** Each row shows the mean Δ ROC-AUC (×100) between ModernMolBERT-base and one baseline (positive = ModernMolBERT-base higher), with 95% bootstrap confidence intervals ($B = 10{,}000$ resamples) and the number of datasets on which ModernMolBERT-base scores higher. The dashed line marks zero. The intervals for SELFormer and ChemBERTa-2 lie entirely above zero, that for MoLFormer entirely below, and that for ECFP4 spans zero. Numerical values are given in Table 7 and full per-dataset results in Table 6.

Model	TDC-ADME	TDC-Tox	TDC-HTS	MoleculeNet	Overall
ECFP4	81.1	85.0	**68.6**	74.8	78.9
ChemBERTa-2 (MLM)	77.3	73.1	66.3	67.2	72.9
SELFormer	75.1	74.6	61.1	68.8	72.1
MoLFormer	**81.5**	**85.2**	66.8	**77.6**	**79.8**
ModernMolBERT-small	79.6	83.7	64.8	73.2	77.3
ModernMolBERT-base	80.7	83.8	64.4	70.7	77.4

Table 2: Mean ROC-AUC (×100) on the 25-dataset benchmark of Praski et al. [19], broken down by task group. Each entry averages per-dataset ROC-AUC using the best cross-validated downstream head (ridge / random forest / kNN); for multi-endpoint datasets, endpoint scores are averaged before aggregation. *Overall* is the unweighted mean across completed benchmark datasets (25 for ModernMolBERT-small; 24 for ModernMolBERT-base, which lacks Tox21; see Table 6). **Bold** marks the best value per column.

On MoleculeNet, MoLFormer shows its clearest advantage (77.6 vs. ModernMolBERT-base 70.7), consistent with its much larger and more chemically diverse pretraining corpus. This pattern is consistent with frozen learned embeddings transferring best to tasks governed by global physicochemical character rather than rare substructure recognition, an interpretation in line with broader analyses of when learned molecular representations help [29, 30].

Relative to ECFP4, ModernMolBERT approaches but does not surpass the fingerprint baseline. ModernMolBERT-base exceeds ECFP4 on 7 of the 23 jointly evaluated datasets, with a mean Δ ROC-AUC of -1.0 and a 95% bootstrap confidence interval of $[-2.5, +0.5]$ (Table 7); the interval spans zero, consistent with neither a reliable advantage nor a reliable deficit at this scale. ModernMolBERT-small wins on 5 of 25. As such, ModernMolBERT is approaching, rather than surpassing, ECFP4 under the frozen-embedding protocol, consistent with the continued strength of circular fingerprints in low-data property prediction.

The comparison most diagnostic of ModernMolBERT's design is SELFormer, the closest prior SELFIES encoder (a RoBERTa-era backbone on the same representation). Here ModernMolBERT improves clearly and consistently: ModernMolBERT-base exceeds SELFormer on 21 of 23 jointly evaluated datasets (mean Δ ROC-AUC +5.4; 95% CI $[+3.9, +7.1]$; Table 7), and ModernMolBERT-small on 24 of 25. Because both models consume SELFIES inputs, this gap is consistent with the combined benefit of ModernBERT-era architectural choices, APE tokenisation, and differences in the training recipe; matched single-axis ablations would be required to assign the gain to individual components. ModernMolBERT likewise exceeds the SMILES-based ChemBERTa-2 by 4.6 points on 23 jointly evaluated datasets (mean Δ +4.6; 95% CI $[+1.6, +7.9]$). It does not reach MoLFormer, which is pre-trained on roughly 400$\times$ more molecules (mean Δ -1.7; 95% CI $[-2.6, -0.7]$).

A qualitative view of the learned representation is given in Figure 6: PaCMAP projections of the frozen ModernMolBERT-base embeddings organise molecules along smooth, chemically interpretable gradients in physicochemical descriptors spanning both whole-molecule properties (lipophilicity, drug-likeness, polarity, size) and more local structural counts (flexibility, aromaticity), indicating that the embedding space encodes chemically meaningful structure even though it is never given these properties as training signal. We read this only qualitatively: the smooth arrangement shows that neighbouring embeddings tend to correspond to chemically similar molecules, which is the property that makes the embeddings useful for the similarity-driven downstream heads above, and we draw no quantitative conclusions from the projected coordinates themselves.

5.2 Ablation Studies

We ablate two pretraining axes — the MLM masking strategy and model size — on validation MLM metrics (Figure 7; full heatmaps in Figure 10) and on downstream ROC-AUC (Figure 9). The three design choices shared by all released models (the SELFIES representation, the APE tokeniser, and the ModernBERT backbone) are not isolated here by matched single-axis ablations; they are instead probed jointly through the external baselines of Section 5.1. The SELFormer comparison, which shares the SELFIES representation and dedicated tokeniser and representation ablations are left for future work.

On the validation MLM objective, the choices separate more clearly: for the small model, span masking at 2×10^{-4} is strongest and heteroatom-biased span masking consistently weakest; for the base model, 2×10^{-4} and 4×10^{-4} behave similarly while 8×10^{-4} diverges. These differences largely do not carry through to downstream property prediction: relative

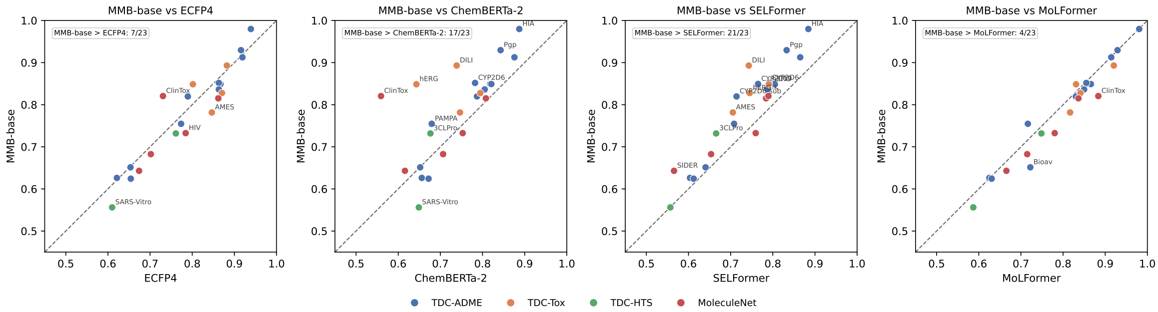

Figure 4: **Per-task comparison of ModernMolBERT-base against the four baseline embedders.** Each panel is a paired scatter of held-out test ROC-AUC across the benchmark tasks (one point per task, best cross-validated downstream head per task), with ModernMolBERT-base on the y-axis and one baseline on the x-axis: ECFP4, ChemBERTa-2 (MLM), SELFormer, and MoLFormer. The dashed line is the identity $y = x$; points above it indicate tasks where ModernMolBERT-base scores higher. Points are coloured by task group (TDC-ADME, TDC-Tox, TDC-HTS, MoleculeNet), tasks with a gap > 0.05 ROC-AUC are labelled, and each inset gives the number of tasks on which ModernMolBERT-base wins. All encoders are evaluated as frozen embedders under an identical protocol; Tox21 is omitted for ModernMolBERT-base (Table 6).

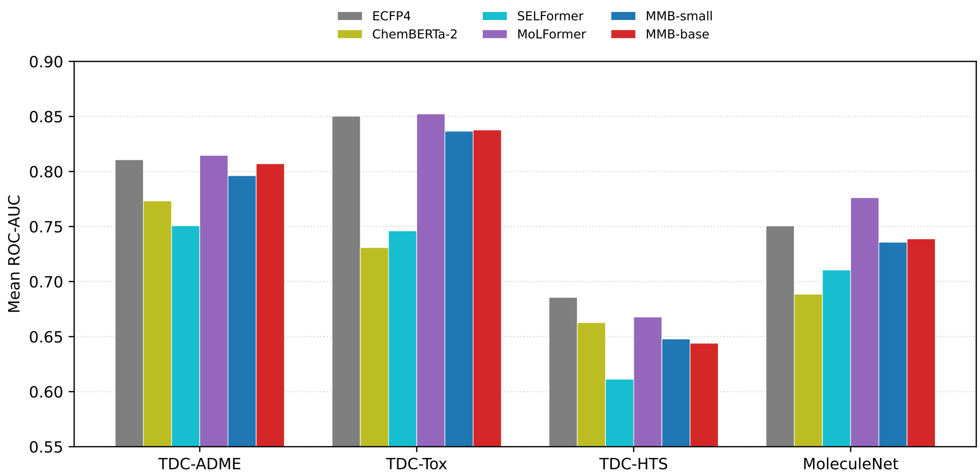

Figure 5: **Mean test ROC-AUC by task group.** Unweighted mean held-out ROC-AUC within each of the four task groups (TDC-ADME, 12 tasks; TDC-Tox, 4; TDC-HTS, 2; MoleculeNet, 7) for ECFP4, ChemBERTa-2 (MLM), SELFormer, MoLFormer, and the two released ModernMolBERT variants, all evaluated as frozen embedders. Group means for the masking-strategy ablation variants are omitted here for clarity and reported in Table 6.

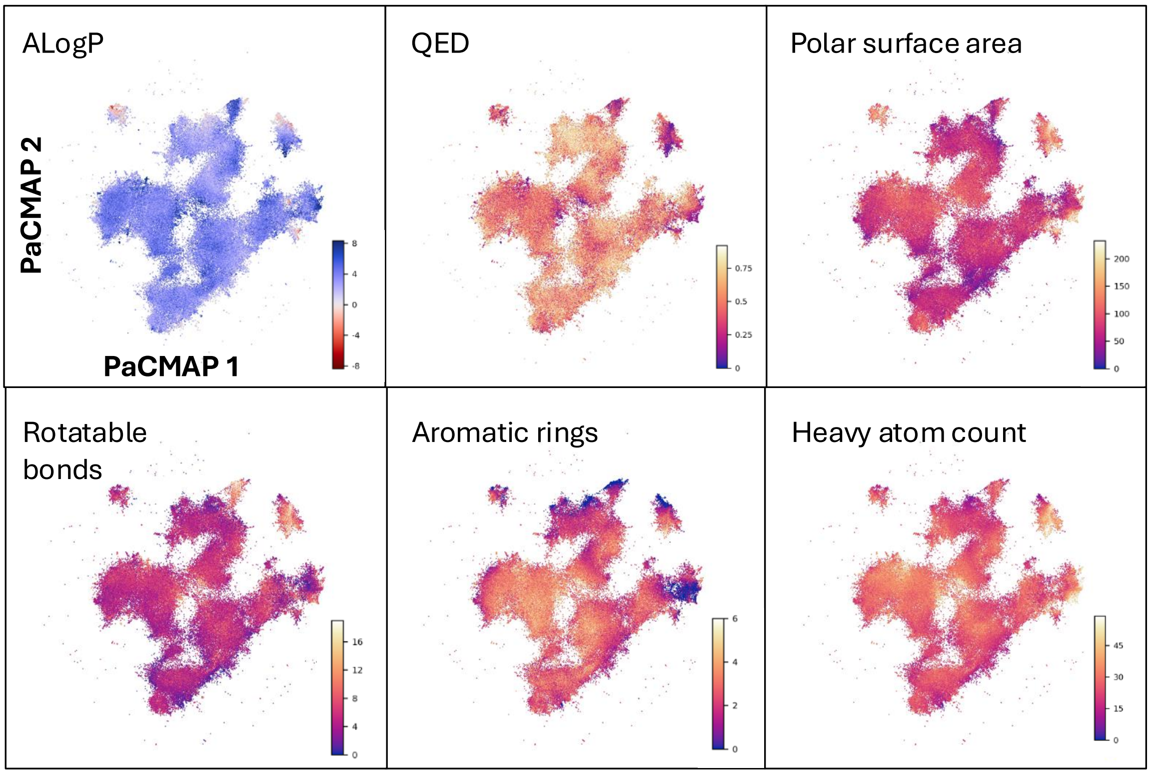

Figure 6: **Chemical property gradients across the ModernMolBERT embedding space.** PaCMAP projection [39] of frozen ModernMolBERT molecular embeddings, coloured by selected physicochemical properties from ChEMBL 36. Each point represents one molecule; colour indicates the corresponding property value: ALogP, the quantitative estimate of drug-likeness (QED), polar surface area, rotatable bonds, aromatic rings, and heavy atom count. PaCMAP is used solely for two-dimensional visualisation of the high-dimensional embeddings; the absolute axis coordinates are not chemically meaningful, and only relative neighbourhoods and broad spatial trends should be interpreted.

to the released small model, the small-to-base size increase changes mean ROC-AUC by +0.1 point (across 24 jointly evaluated datasets), span masking by +0.1 (23 datasets), and heteroatom-biased span masking by −0.7 (22 datasets), all within the dataset-to-dataset scatter. The heteroatom-biased deficit is a genuine negative result: enriching masking around heteroatom tokens is insufficient to improve frozen embeddings, suggesting the gap to the explicit functional-group masking of Peng et al. [18] matters in practice.

Frozen-embedding quality is therefore largely insensitive to masking strategy and to the small-to-base size increase. This supports releasing the simplest configuration (standard masking) as the default and is consistent with the larger gains over SELFormer (Section 5.1) arising from the design choices shared by all released models. The corresponding per-task paired scatters isolating each pretraining axis are provided in Figure 9 (Section C).

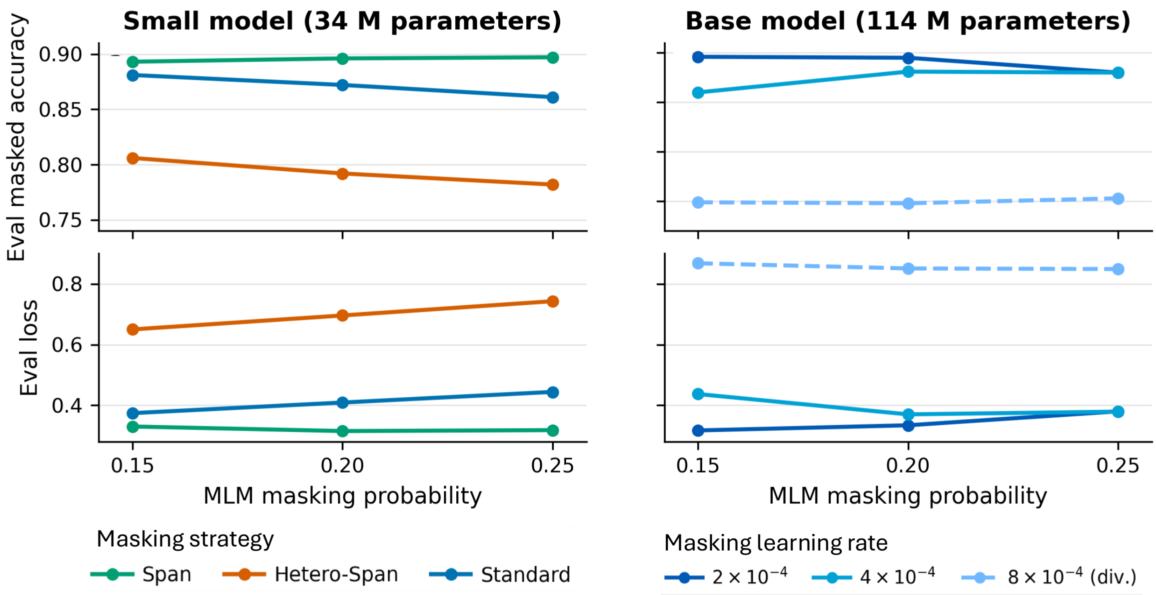

Figure 7: **Primary sweep of MLM masking probability, masking strategy, and learning rate.** Validation performance across MLM masking probabilities for the small ModernMolBERT (∼34 M parameters) and base ModernMolBERT (∼114 M parameters) models. **(a,b)** evaluation masked accuracy (higher is better); **(c,d)** evaluation loss (lower is better); the x-axis is the MLM masking probability throughout. For the small model, line colour denotes masking strategy (span, hetero-span, standard); for the base model, all runs use standard masking and line colour denotes peak learning rate (2×10^{-4}, 4×10^{-4}, 8×10^{-4}). Shaded bands give the min–max range across runs and the asterisk marks the run that diverged during training. Full heatmaps across all conditions are provided in Figure 10.

6 Discussion

6.1 Scope and interpretation

ModernMolBERT is designed around a narrow but practical question: whether a compact encoder-only model can provide useful frozen molecular embeddings when trained on chemically valid SELFIES strings with a SELFIES-aware APE tokeniser. This scope differs from large generative chemical language models, multimodal assay–molecule models, and graph or 3D molecular encoders. The resulting comparison is therefore most informative when interpreted as a test of frozen, string-based molecular representation learning rather than as a claim about all molecular foundation models. ModernMolBERT is used here purely as an encoder, not as a generative model; we therefore do not sample or reconstruct molecules from it. We note, however, that because both inputs and the MLM head operate over the SELFIES vocabulary, any SELFIES sequence the model predicts decodes to a chemically valid molecule by construction, so a generative use of the architecture would inherit SELFIES' validity guarantee — a direction we leave to future work.

Every number in this paper should be read as a *frozen-embedding* result. No encoder is fine-tuned downstream; each produces a fixed mean-pooled embedding that is passed to lightweight classifiers. This protocol isolates representation transferability and mirrors fingerprint-based workflows, where fixed molecular features are precomputed once and reused across many endpoints. The fine-tuned behaviour of ModernMolBERT is left to future work (Section 6.4).

6.2 Advances over prior string models

Within this frozen-feature setting, the clearest advance is over the closest prior SELFIES encoder: holding the molecular representation fixed, ModernMolBERT improves on SELFormer by roughly five ROC-AUC points and is similarly ahead of the closest-scale SMILES MLM, ChemBERTa-2 (Section 5.1). Because the SELFormer comparison shares the SELFIES representation, this gain is consistent with benefits from the ModernBERT backbone and APE tokeniser, though matched single-axis ablations would be required to isolate each component's contribution. We do not overstate the result: ModernMolBERT is competitive with, but does not surpass, the strong ECFP4 fingerprint baseline and trails MoLFormer, which is pre-trained on roughly 400× more molecules [10]. The contribution is therefore a compact, reproducible SELFIES encoder that approaches fixed-fingerprint quality while substantially improving on prior string-based language models at comparable scale.

A practical advantage of ModernMolBERT is accessibility. ModernMolBERT is released in the standard Hugging Face Transformers format and can be used with the current Python ecosystem without model-specific runtime dependencies, legacy Python versions, custom inference kernels, or external chemistry toolkits at embedding time once inputs are provided as SELFIES strings. Frozen embeddings can therefore be extracted with a few lines of standard Transformers code, making the model easier to incorporate into existing screening, retrieval, and benchmark pipelines than models requiring specialised environments. Likewise, the standalone tokeniser is straightforward to use in a similar fashion.

ModernMolBERT therefore occupies a practical trade-off point rather than a pure performance extreme. Larger or more specialised models remain preferable when maximum benchmark accuracy is the only objective, but many molecular-embedding workflows require representations that are adequate, reproducible, compact, and easy to deploy. In this setting, ModernMolBERT is a reasonable default: it approaches ECFP4, improves over comparable pretrained string encoders, and exposes frozen embeddings through the standard Hugging Face Transformers interface without specialised inference infrastructure.

6.3 Limitations

ModernMolBERT is pre-trained exclusively on ChEMBL 36, a corpus dominated by drug-like, Lipinski-compliant compounds. Compared with MoLFormer, which draws from 1.1B PubChem [40] and ZINC [41] molecules, ModernMolBERT's training distribution is narrower; generalisation to natural products, agrochemicals, peptides, macrocycles, or other under-represented chemical spaces may be limited. Pretraining uses canonical-SMILES-derived SELFIES strings only; we do not apply randomised SMILES augmentation or multiple-enumeration strategies, which have been explored as a means of improving representation diversity in prior molecular language models [21, 42]. Like all sequential string models, ModernMolBERT has no access to explicit 3D conformational information; graph- and geometry-based models such as Uni-Mol [28] and GEM [43] address this at the cost of greater architectural complexity. The SELFIES-adapted APE tokeniser forms merge candidates from sequence-adjacent SELFIES primitives rather than reconstructing the molecular graph; merge boundaries therefore do not necessarily align with graph-level bond adjacency. Our evaluation is restricted to binary classification tasks scored by ROC-AUC, following the benchmark of Praski et al. [19]; performance on regression tasks, generative objectives, or out-of-distribution scaffolds is not assessed here. A further caveat concerns possible train–benchmark overlap: both the ChEMBL 36 pretraining corpus and several of the benchmark datasets are derived from overlapping medicinal-chemistry sources, and we do not explicitly deduplicate benchmark structures against the pretraining set. Some benchmark molecules or close analogues may therefore also appear during pretraining, which could modestly inflate absolute scores for all pretrained encoders evaluated here; the relative comparisons between encoders are less affected, since every pretrained baseline is exposed to similar public chemical space. Finally, all models are evaluated as frozen encoders, which may not reflect relative performance under full fine-tuning; frozen evaluation may favour representations with broader but shallower chemical coverage. More generally, systematic comparisons of molecular property prediction models show that representation learning gains are highly context-dependent and are affected by data regime, split strategy, and activity cliffs [29, 30]. These considerations are especially relevant for frozen-feature evaluations, where the representation must transfer without task-specific adaptation.

6.4 Future Work

Several extensions follow naturally from the present design, all within the MLM-only, string-encoder scope of this work. The most direct is broader pretraining beyond ChEMBL, including PubChem, ZINC, Enamine, or other sources that better cover natural products, fragments, reagents, and non-drug-like chemistry. Pretraining ModernMolBERT on the same corpora used by the baselines—for example the PubChem/ZINC mixture behind MoLFormer—would further allow the architecture and tokeniser to be compared at matched data, separating the contribution of pretraining scale from that of model design. A closely related question is how ModernMolBERT scales: the present study fixes a compact small/base pair and a 30,000-step budget, and systematically increasing model size, corpus size, and training compute would establish whether the remaining gap to large-scale SMILES models such as MoLFormer narrows with scale. A second direction is matched single-axis ablations of the tokeniser and backbone choices—substituting APE with standard BPE or the ModernBERT backbone with a RoBERTa-era encoder—to isolate each component's contribution to the gains over SELFormer. A third is to explore SELFIES encoders as molecular indexing models for chemical search, analogue retrieval, and dataset auditing, where deterministic valid string encodings and compact frozen embeddings are operationally attractive.

7 Conclusion

We presented ModernMolBERT, a family of compact encoder-only transformer models for SELFIES molecular language modelling. The models combine the ModernBERT encoder architecture with a SELFIES-adapted APE tokeniser and are pre-trained from scratch on a curated ChEMBL 36 corpus using the masked language modelling objective.

On 24 completed benchmark datasets evaluated under the frozen embedding protocol, ModernMolBERT-base achieves a mean ROC-AUC of 77.4. This is competitive with the strong ECFP4 fingerprint baseline (78.9) and trails the much larger MoLFormer (79.8), while clearly exceeding the closest prior SELFIES and SMILES string encoders, SELFormer (72.1) and ChemBERTa-2 (72.9), by four to five ROC-AUC points. Internal ablations further show that downstream embedding quality is largely insensitive to masking strategy and to the small-to-base size increase, so the gains over prior string models are consistent with the shared SELFIES, APE, and ModernBERT design choices. These results suggest that a modern SELFIES encoder with an atom-preserving APE tokeniser can approach fingerprint-level frozen-feature performance while substantially improving on earlier string-based language models at comparable scale.

The central motivation is practical: molecular string encoders should be valid by construction, tokenised in chemically meaningful units, and built on modern efficient transformer components rather than inherited BERT-era backbones. By evaluating ModernMolBERT as a frozen embedder against ECFP4 and representative pre-trained molecular language models, we position SELFIES-based ModernBERT encoders as a reproducible and computationally manageable foundation for molecular representation learning.

Data and Code Availability

Training code and evaluation scripts are available at https://github.com/HauserGroup/ModernMolBERT. Model weights and the APE tokeniser are available on Hugging Face at https://huggingface.co/HauserGroup. ChEMBL 36 is publicly available at https://www.ebi.ac.uk/chembl/.

CRediT authorship contribution statement

J.S.M. Writing – original draft, Methodology, Formal analysis, Conceptualization S.A. Methodology. X.L. Methodology. A.S.H. Methodology, Funding acquisition, Conceptualization.

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Competing Interests

The authors declare no competing interests.

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Appendix A Training Hyper-parameters

Hyper-parameter	**ModernMolBERT-small**	**ModernMolBERT-base**
Parameters	34.1M	114.3M
Hidden size	512	768
Transformer layers	8	12
Attention heads	8	12
FFN intermediate size	2 048	3 072
Global attn. every n layers	3	3
Local attention window	128	128

Table 3: Hyper-parameters for the two ModernMolBERT variants. Both use the ModernBERT backbone with alternating local/global attention (global every 3 layers; local window of 128 tokens).

Hyperparameter	**Small**	**Base**
Fixed		
Max training steps	30,000	30,000
Max sequence length	128	128
Effective batch size	256	256
Per-device batch size	256	64
Gradient accumulation	1	4
Weight decay	0.01	0.01
Gradient clip norm	1.0	1.0
Precision	bf16	bf16
Seed	42	42
Swept (3×3 grid, 9 runs each)		
MLM masking rate	$\{0.15,\ 0.20,\ 0.25\}$	
Peak learning rate	$\{10^{-4},\ 2\times10^{-4},\ 4\times10^{-4}\}$	$\{2\times10^{-4},\ 4\times10^{-4},\ 8\times10^{-4}\}$
LR warmup steps	1,500	2,000

Table 4: Pretraining hyperparameters for both ModernMolBERT variants. Swept values are shown in braces. The released checkpoints use the stable standard-masking configuration (4×10^{-4} for small, 2×10^{-4} for base); the full sweep is analysed in Section 5.2.

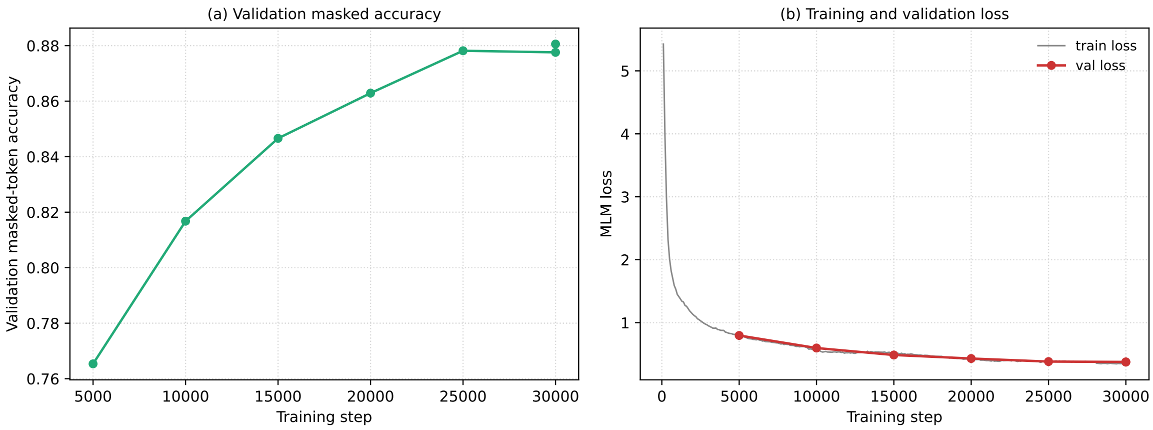

Figure 8: **Pretraining convergence of the released small ModernMolBERT.** Validation curves for the small model (standard masking, 4×10^{-4}) over 30,000 training steps. **(a)** Validation masked-token accuracy. **(b)** Training loss (dense) and validation MLM loss (markers). Both curves plateau smoothly, confirming stable convergence within the fixed 30,000-step budget. Per-step logs for the base model were not retained; its final validation metrics are reported in the sweep (Figure 7).

Appendix B Tokeniser Details

Tokeniser property	**Value**
Vocabulary	
Training molecules	2000000
Base vocabulary (APE merges)	589
Added SELFIES primitives	42
Final vocabulary	631
Sequence statistics (10 000 held-out ChEMBL 36 molecules)	
Median length (tokens)	24
Mean length (tokens)	25.7
95th percentile	43
99th percentile	56
Coverage	
Truncation at 128 tokens	<0.1%
Unknown-token rate	0.0%

Table 5: Summary statistics for the SELFIES-adapted APE tokeniser. The base vocabulary is learned by APE merge training on 2M randomly sampled ChEMBL 36 SELFIES strings; 42 additional SELFIES primitives observed in downstream benchmark molecules are appended after merge learning but before model pretraining, giving a final vocabulary of 631 tokens on which all model weights are initialised and trained. Sequence statistics and coverage are evaluated on 10 000 held-out ChEMBL 36 molecules.

The final tokeniser is trained on 2M randomly sampled ChEMBL 36 SELFIES strings with `max_merge_pieces` = 2 and `min_freq_for_merge` = 3000, yielding a base vocabulary of 589 tokens. An additional 42 SELFIES primitives observed in the benchmark datasets (minimum frequency ≥ 10) are appended after APE merge learning but before model pretraining, giving a final vocabulary of 631 tokens on which all model weights were initialised and trained from the outset. Special tokens are assigned fixed ids: `[BOS]` $= 0$, `[PAD]` $= 1$, `[EOS]` $= 2$, `[UNK]` $= 3$, `[MASK]` $= 4$. Evaluated on 10 000 held-out ChEMBL 36 molecules, the tokeniser achieves an unknown-token rate of 0.0%, a median sequence length of 24 tokens (mean 25.7; 95th percentile 43; 99th percentile 56), negligible truncation at the model context length of 128 tokens (<0.1%), and zero truncations at 256 tokens.

Appendix C Additional Benchmark Results

This appendix reports per-task benchmark results, together with auxiliary analyses used to assess robustness of the frozen-feature evaluation. Table 6 gives the full per-task test ROC-AUC for every model, including the span- and heteroatom-span masking variants of ModernMolBERT-small. Results not available for a given model–dataset combination are marked "–": Tox21 for ModernMolBERT-base; HIV and MUV for the span-masking variant; and MUV, SIDER, and Tox21 for the heteroatom-span variant.

Task	**ECFP4**	**ChBa-2**	**SELF.**	**MoLF.**	**MMB-s**	**MMB-b**	**MMB-sp**	**MMB-h**
TDC – ADME								
Bioavailability	65.3	65.2	64.1	72.2	63.1	65.1	67.6	71.1
HIA	93.9	88.7	88.4	98.1	97.8	98.0	98.1	97.9
Pgp	91.6	84.3	83.3	92.9	91.5	93.0	90.6	92.0
PAMPA	77.4	67.9	70.8	71.6	73.9	75.5	72.2	75.5
CYP1A2	91.9	87.6	86.5	91.4	90.4	91.3	89.9	90.9
CYP2C19	86.8	81.8	76.5	85.5	84.2	84.9	83.1	84.4
CYP2C9	86.7	82.1	80.5	86.6	84.9	84.9	84.6	85.3
CYP2D6	86.3	78.2	79.1	85.5	83.3	85.2	83.6	85.1
CYP3A4	86.3	80.5	78.7	85.0	83.1	83.6	81.9	83.3
CYP2C9 (substrate)	62.1	65.6	60.4	62.5	62.0	62.6	60.7	59.4
CYP2D6 (substrate)	79.0	78.7	71.4	83.0	79.2	82.0	78.8	78.8
CYP3A4 (substrate)	65.4	67.2	61.2	63.1	62.1	62.4	65.0	66.9
TDC – Toxicity								
AMES	84.7	74.7	70.5	81.7	78.3	78.2	78.2	78.2
DILI	88.2	73.9	74.3	92.0	92.3	89.3	93.2	90.2
hERG	80.2	64.3	79.1	83.1	81.6	84.9	80.5	80.2
hERG (Karim)	87.1	79.5	74.5	84.1	82.4	82.8	81.4	82.2
TDC – HTS								
SARS-CoV-2 3CLPro	76.1	67.6	66.6	74.9	70.7	73.2	73.0	71.9
SARS-CoV-2 (Vitro)	61.0	64.9	55.7	58.7	58.9	55.6	54.6	56.5
MoleculeNet								
BACE	86.2	80.8	78.4	83.7	78.4	81.5	78.6	80.1
BBBP	70.2	70.7	65.4	71.5	69.3	68.3	69.3	68.3
ClinTox	73.1	55.9	79.0	88.4	79.2	82.1	81.6	83.2
HIV	78.4	75.3	75.9	78.0	76.2	73.3	–	75.5
MUV	72.4	56.6	57.7	79.8	70.5	54.7	–	–
SIDER	67.4	61.6	56.6	66.6	64.7	64.3	63.5	–
Tox21	75.9	69.4	69.0	75.0	73.9	–	73.6	–

Table 6: Per-task test ROC-AUC (×100) for all models on the 25-task benchmark. *MMB-s* = ModernMolBERT-small (standard), *MMB-b* = ModernMolBERT-base, *MMB-sp* = small span masking, *MMB-h* = small hetero-span masking. "–" marks evaluations not yet run.

Model A	Model B	n	Wins	Ties	Losses	Mean Δ	95 % CI
MMB-base	SELFormer	23	21	0	2	+5.43	[+3.9, +7.1]
MMB-base	ChemBERTa-2	23	17	0	6	+4.55	[+1.6, +7.9]
MMB-base	ECFP4	23	7	0	16	-1.02	[−2.5, +0.5]
MMB-base	MoLFormer	23	4	0	19	-1.67	[−2.6, −0.7]

Table 7: Paired bootstrap confidence intervals (95 %, $B = 10{,}000$ resamples) on mean Δ ROC-AUC (×100) between ModernMolBERT-base and each baseline, computed over the tasks where both models have a result. *Wins/Ties/Losses* count tasks where ModernMolBERT-base is above, equal to, or below the baseline. A positive mean Δ and CI entirely above zero indicates a consistent advantage for ModernMolBERT-base.

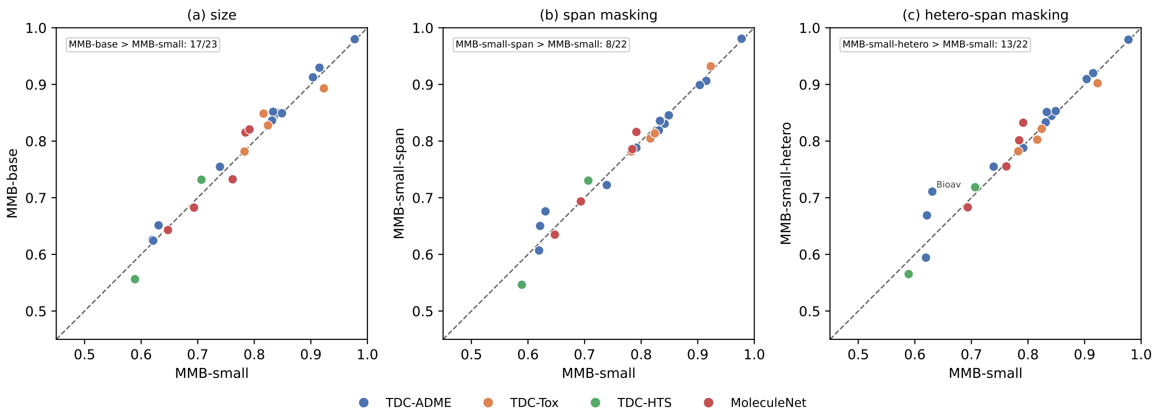

Figure 9: **Internal comparison of the four ModernMolBERT checkpoints on downstream ROC-AUC.** Paired scatter of held-out test ROC-AUC across the benchmark tasks (one point per task, best cross-validated downstream head), isolating each pretraining design axis against the released small model (standard masking, y-axis throughout except panel a). **(a)** ModernMolBERT-small versus ModernMolBERT-base, isolating model size. **(b)** ModernMolBERT-small versus the span-masking variant. **(c)** ModernMolBERT-small versus the heteroatom-biased span-masking variant. The dashed line is the identity $y = x$, points are coloured by task group, and each inset reports the number of tasks on which the y-axis model wins. Panel a is evaluated on 24 jointly evaluated datasets (ModernMolBERT-base lacks Tox21) and panels b and c only on the tasks available for the respective variant (23 and 22 of 25; see Table 6).

Appendix D Molecular Embedding Models

Table 8 lists the non-graph molecular embedding models evaluated in the benchmark of Praski et al. [19]. Graph-based models (R-MAT, MAT, GROVER, GIN, MolR, GraphMVP, Uni-Mol, Uni-Mol2, GEM, GraphFP) are excluded here as they fall outside the scope of ModernMolBERT. Table 6 of Praski et al. [19] provides the full benchmark results across all 25 methods, including graph-based architectures.

Model	**Input**	**Params (M)**	**Dim.**	**Year**	**Corpus**	**Size (M)**
Atom Pair	Fingerprint	—	2,048	1985	—	—
TT	Fingerprint	—	2,048	1987	—	—
ECFP	Fingerprint	—	2,048	2010	—	—
Mol2Vec	Substructure	n.r.	300	2018	ChEMBL	n.r.
CDDD	SMILES	n.r.	512	2019	n.r.	n.r.
MolBERT	SMILES	85	768	2020	ChEMBL	2
ChemBERTa	SMILES	5–46[a]	384	2020	PubChem	77
ChemGPT	SMILES/SELFIES	5	128	2022	PubChem	10
Chemformer	SMILES	45	512	2022	ZINC-15	100
MoLFormer	SMILES	46.8	768	2022	PubChem + ZINC	>1,100
CLAMP	Fingerprint/RDKit	n.r.	768	2023	PubChem assays	143 meas.
COATI	SMILES/3D	n.r.	256	2023	n.r.	n.r.
SELFormer	SELFIES	87	768	2023	ChEMBL	2
SimSon	SMILES	n.r.	512	2025	n.r.	n.r.
ChemFM	SMILES	3,000	3,072	2025	UniChem	178
ModernMolBERT	SELFIES	34/114	512/768	2026	ChEMBL 36	2.4

Table 8: Non-graph molecular embedding models considered in this work, following the benchmark selection of Praski et al. [19]. Graph-based architectures are excluded; see Table 6 of that work for the full model comparison. *Params*: trainable parameters in millions; *Size*: pretraining corpus size in millions unless otherwise stated; n.r. = not reported clearly in the paper, model card, or repository; "—" = not applicable. [a]For ChemBERTa, 5M/10M/77M denote pretraining corpus sizes, not parameter counts; reported architecture searches span approximately 5–46M parameters.

Appendix E Pretraining Hyperparameter Heatmaps

This appendix reports the full hyperparameter sweep underlying the masking and learning-rate selections discussed in Section 5.2, as heatmaps over all masking probabilities, strategies, and learning rates evaluated.

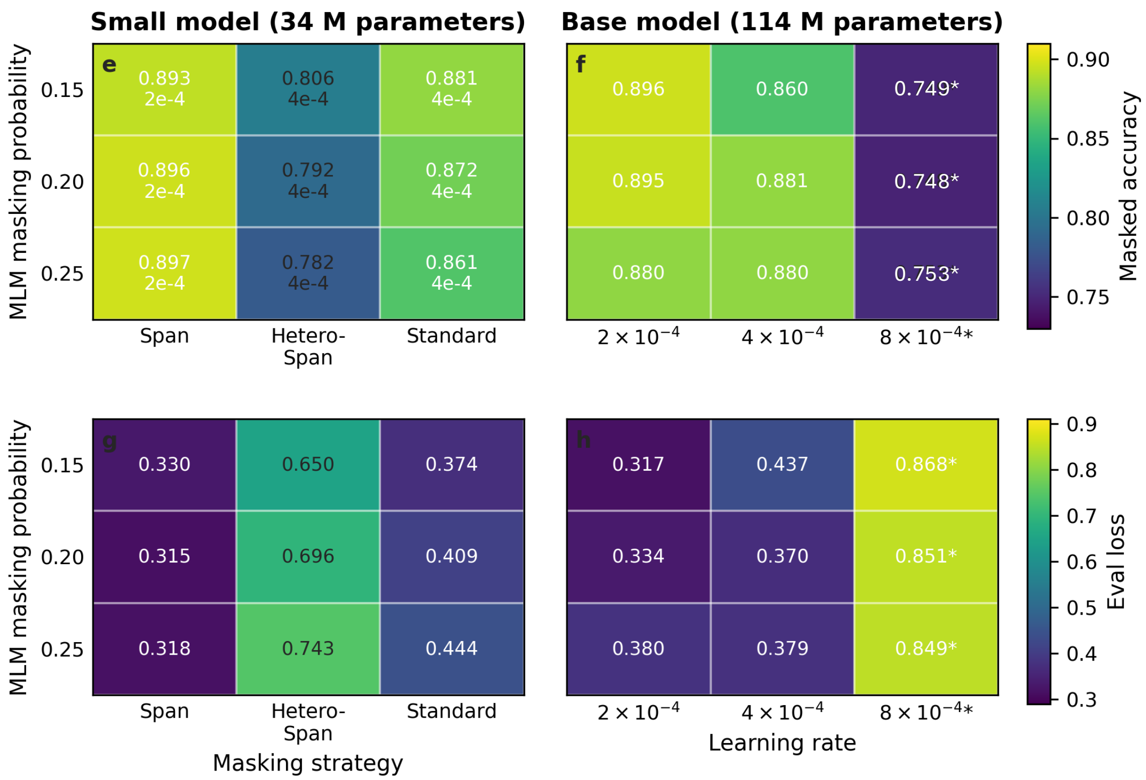

Figure 10: **Masked language modelling sweep across masking strategies, masking probabilities, and learning rates.** Heatmaps show validation performance for the small ModernMolBERT model (∼34 M parameters) and the base ModernMolBERT model (∼114 M parameters). Panels **e** and **f** show masked-token accuracy (higher is better). Panels **g** and **h** show evaluation loss (lower is better). For the small model, columns compare masking strategies: span masking, hetero-span masking, and standard token masking. For the base model, columns compare peak learning rates under standard masking: 2×10^{-4}, 4×10^{-4}, and 8×10^{-4}. Rows give the MLM masking probability: 0.15, 0.20, and 0.25. For the small model, the second line in each cell reports the learning rate used; asterisks denote runs that diverged during training. On validation MLM metrics, the small model performs best with span masking at a learning rate of 2×10^{-4}, with accuracy increasing slightly as masking probability increases. Hetero-span masking is consistently worse for both accuracy and loss. For the base model, 2×10^{-4} and 4×10^{-4} perform similarly; 8×10^{-4} produces unstable training with superficially moderate final accuracy but elevated loss. Line-plot views of panels a–d are shown in the main text (Figure 7).